AVINASH KARKADA ASHOK

Biotechnology | Bioinformatics/Computational Biology

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Bioinformatics and computational biology professional with 4+ years of experience across NGS workflows, immune profiling, structural biology, and applied data analysis. Built and maintained analysis pipelines, automated workflows, and supported reproducible research using Python, R, Bash, Snakemake, other open source and proprietary tools, and Docker on Linux/HPC and cloud systems. Experience spans both academic research and biotech environments, with 12+ peer-reviewed publications.

EDUCATION

M.S. Bioinformatics
Johns Hopkins University
Aug 2024 – May 2026 | Baltimore, USA

B.E. Biotechnology
Visvesvaraya Technological University
June 2019 – July 2023 | Karnataka, India
First Class with Distinction

EXPERIENCE

Bioinformatics Engineer
Johns Hopkins University School of Medicine
Sep 2024 – Present | Baltimore, USA

- Engineer and maintain computational pipelines for InFlux, a highly multiplexed NGS-based influenza A neutralization assay that generates strain-specific neutralization measures per sample, including pipeline validation and QC workflows to ensure data quality across diverse patient cohorts in **Dr. Benjamin H. Larman's** lab.
- Independently lead a **PhIP-Seq** antibody-profiling project analyzing several hundred serum samples investigating microbial dysbiosis and humoral immunity in early-onset, late-onset colorectal cancer, and polyp/healthy control cohorts. Anticipated first-author manuscript in progress.

Summer Bioinformatics Intern
Infinity Bio, Inc.
June 2025 – August 2025 | Baltimore, USA

- Benchmarked and optimized NGS-based **MIPSA** analysis pipelines by evaluating alignment strategies, improving alignment quality and I/O handling while reducing overall runtime and resource usage.
- Co-developed EpitopeFindeR 2, a reproducible pipeline integrating FoldDisco-based 3D structural mapping to identify continuous and discontinuous epitopes from library peptide sequences. Pipeline deployed to production for client-facing data portal.
- Built **Latch Bio**-based analysis pipelines including a sequence-space analyzer and visualization module, improving customer experience for EpitopeFindeR 2 users.

Junior Research Fellow
Indian Institute of Science
Nov 2023 – Aug 2024 | Bangalore, India

- Co-first-authored a comparative genomics study decoding mammalian adaptive signatures in clade 2.3.4.4b H5N1 across non-human hosts, identifying convergent mutations that may increase human pandemic risk (published in *Microbiology Spectrum*).
- Demonstrated that neuraminidase stalk length and haemagglutinin glycosylation patterns molecularly bias segment pairing, elucidating reassortment routes that facilitated emergence of highly pathogenic clade 2.3.4.4b A(H5N1).
- Contributed to the discovery and characterization of RHIM motifs in bat RNA viruses including a SARS-CoV-2 Nsp13 RHIM, revealing species-specific ZBP1–RIPK3 cell death signaling (published in *Cell iScience*).

SKILLS

Programming & Platforms

Python | R | Bash | SQL | Java | Git/GitHub | Linux | HPC/SLURM | AWS (EC2, S3, GPU) | Docker | Singularity | Conda/Mamba | Jupyter | RStudio | Quarto | Shiny

NGS & Omics Analysis

PhIP-Seq | RNA-seq | scRNA-seq | ChIP-seq | WGS | WES | MIPSA | Bowtie2 | BWA | Minimap2 | STAR | HISAT2 | Kallisto | Salmon | SAMtools | BCFtools | BEDTools | GATK | Variant Calling | FastQC | Trimmomatic | Cutadapt | SRA Toolkit | IGV | DESeq2 | edgeR | limma | Seurat | Scanpy | Bioconductor | Biopython | BLAST | CD-HIT | HMMER | MAFFT | Epitope Mapping | Differential Expression | Enrichment Analysis | Pathway Analysis | Network Analysis | Clustering | IQ-TREE

Workflow & Reproducibility

Snakemake | Latch Bio | Docker | Singularity | Git/GitHub

ML, Statistics & Visualization

scikit-learn | XGBoost | SVM | LASSO | Random Forests | PCA | UMAP | t-SNE | pandas | NumPy | SciPy | matplotlib | plotly | ggplot2 | dplyr | tidyr | GraphPad Prism | BioRender

Structural Biology & Drug Discovery

GROMACS | AMBER | gmx_MMPBSA | AlphaFold | ESMFold | Rosetta | HADDOCK | AutoDock | AutoDock Vina | Maestro/Schrödinger | RDKit | PyMOL | UCSF ChimeraX | Virtual Screening | ADMET | Protein Structure Prediction

Wet Lab Techniques

Bacterial cell culture | PCR | Gel electrophoresis | SDS-PAGE | Western blot | ELISA | Sample handling & QC

RELEVANT PUBLICATIONS

- **Decoding non-human mammalian adaptive signatures of clade 2.3.4.4b H5N1 to assess its human adaptive potential**
  R. Nataraj, **A. K. Ashok**, A. A. Dey, and S. Kesavardhana
  Microbiology Spectrum.
- **High-throughput screening and molecular dynamics simulations of natural products targeting LuxS/AI-2 system as a novel antibacterial strategy for antibiotic resistance in *Helicobacter pylori***
  A. K. Ashok, T. S. Gnanasekaran, H. S. S. Kumar, K. Srikanth, N. Prakash, and P. Gollapalli
  Journal of Biomolecular Structure and Dynamics.
- **Bat RNA viruses employ viral RHIMs orchestrating species-specific cell death programs linked to Z-RNA sensing and ZBP1–RIPK3 signaling**
  S. Mishra, D. Jain, A. A. Dey, S. Nagaraja, M. Srivastava, O. Khatun, K. Balamurugan, M. Anand, **A. K. Ashok**, S. Tripathi, *et al.*
  Cell iScience.
- **Computational-driven discovery of AI-2 quorum sensing inhibitor targeting the $5'$-methylthioadenosine/S-adenosylhomocysteine nucleosidase (MTAN) to combat drug-resistant *Helicobacter pylori***
  M. Kumar, **A. K. Ashok**, T. Bhat, K. Ballamoole, and P. Gollapalli
  Computers in Biology and Medicine.

ACHIEVEMENTS & SERVICE

- Peer reviewer for *Nature Communications*, *Journal of Crohn's and Colitis*, *J. Biomolecular Structure and Dynamics*, and *J. Biological Macromolecules*.
- 12+ peer-reviewed publications as first- or co-author across influenza genomics, viral cell-death signaling, and computational drug discovery.
- Paper presentation at Immunology 2025 (AAI Annual Meeting).
- Awarded Best Paper Presentation (JAIVIK 2022) and Best Undergraduate Thesis.
- Mentored multiple students in practical bioinformatics, molecular dynamics, and NGS analysis.